Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 12: Combining Sets

Union, Intersection, Difference, and Complement

Chartrand & Zhang, Section 2.2

Name: _____ Date: _____

Where we are

Friday you learned what a set is. Today you learn how to build new sets from old ones — and you will find you already know how, because **every set operation is defined by a logical connective from Week 1**.

By the end of today you should be able to

1. Compute $A \cap B$, $A \cup B$, $A - B$, $A \oplus B$, and $\overline{A}$.
2. Write each operation's membership condition using $\wedge$, $\vee$, and $\sim$.
3. Build a membership table and recognise it as a truth table.

1. Intersection and union

Definitions 2.19 and 2.20

$$A \cap B = \{x : x \in A \textbf{ and } x \in B\} \qquad A \cup B = \{x : x \in A \textbf{ or } x \in B\}$$

Which "or"?

The "or" in the definition of union is the **inclusive or**. An element of $A \cup B$ may belong to A, to B, or to both.
Week 1 spent a whole class on this. Here is where it pays: if $\cup$ used exclusive or, $A \cap B$ would not be a subset of $A \cup B$ — and it is.

Example 2.21. $C = \{1, 2, 4, 5\}$ and $D = \{1, 3, 5\}$.

$$C \cap D = \{1, 5\} \qquad C \cup D = \{1, 2, 3, 4, 5\}$$

Definition 2.25 — disjoint

A and B are **disjoint** if $A \cap B = \emptyset$. A collection is **pairwise disjoint** if every two distinct sets in it are disjoint.

2. Difference, symmetric difference, complement

Definitions 2.28, 2.29, 2.32

$$A - B = \{x : x \in A \text{ and } x \notin B\} \qquad A \oplus B = (A - B) \cup (B - A)$$
$$\overline{A} = \{x \in U : x \notin A\} = U - A \qquad (U = \text{the universal set})$$

That symbol is not a coincidence

The **symmetric difference** $A \oplus B$ uses the same symbol as *exclusive or* — because it is the same idea. $x \in A \oplus B$ exactly when x is in one of the two sets but not both.

Work these. $U = \{1, \ldots, 8\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$.

$A \cap B$	$\{3, 4\}$	$A - B$	$\{1, 2\}$
$A \cup B$	$\{1, 2, 3, 4, 5, 6\}$	$B - A$	$\{5, 6\}$
$\overline{A}$	$\{5, 6, 7, 8\}$	$A \oplus B$	$\{1, 2, 5, 6\}$

Note that $A - B$ and $B - A$ are different. There is no commutative law for set difference — though $A \oplus B = B \oplus A$ always.

3. The point of today: membership is logic

Every operation is defined by a connective. Fill in the right column.

Statement	Equivalent logical form
$x \in A \cap B$	$(x \in A) \wedge (x \in B)$
$x \in A \cup B$	$(x \in A) \vee (x \in B)$
$x \in \overline{A}$	$\sim(x \in A)$
$x \in A - B$	$(x \in A) \wedge \sim(x \in B)$
$x \in A \oplus B$	$(x \in A) \oplus (x \in B)$

Membership tables

Because membership is logic, you can analyse sets with a **membership table** — a truth table whose rows are the possible ways an element can sit relative to the sets.

$x \in A$	$x \in B$	$x \in A \cap B$	$x \in A \cup B$	$x \in A - B$	$x \in A \oplus B$
T	T	T	T	F	F
T	F	F	T	T	T
F	T	F	T	F	T
F	F	F	F	F	F

Four rows, and each row is one **region of a Venn diagram**. Two sets carve the universe into 4 regions; three sets carve it into 8.

4. Two identities you have already proved

Since $\overline{A}$ is the negation of membership in A:

$$A \cup \overline{A} = U \qquad A \cap \overline{A} = \emptyset$$

Lecture 7, in set clothing

The first says $(x \in A) \vee \sim(x \in A)$, which is true for every x — a tautology. So the set of such x is everything: U.
The second says $(x \in A) \wedge \sim(x \in A)$, false for every x — a contradiction. So no x qualifies: $\emptyset$.
U and $\emptyset$ are the set versions of "always true" and "always false."

Example 2.33. Over $U = \mathbb{Z}$, if E is the set of even integers then $\overline{E}$ is the set of odd integers, and indeed $E \cup \overline{E} = \mathbb{Z}$ and $E \cap \overline{E} = \emptyset$.

5. Venn diagrams — useful, but not proofs

Draw and shade $A - B$, then $\overline{A \cup B}$, on your own paper.

A picture is not an argument

Venn diagrams are excellent for *seeing* what an identity claims. They do not *prove* it — the book says so explicitly about Theorem 2.22.
A shaded picture shows one drawing. A membership table covers every case, and a double-inclusion argument covers every element. Those are proofs. Friday we write them.

Exit ticket

1. $U = \{1, \ldots, 6\}$, $A = \{1, 2, 3\}$, $B = \{3, 4\}$. Give $A \oplus B$ and $\overline{A \cup B}$.

2. Write $x \in A - B$ using only $\wedge$, $\vee$, and $\sim$.

Before Wednesday

Read Section 2.2, pp. 63–68 — Theorem 2.22 and Theorem 2.35. You will recognise both.

Self-check (not collected). Section 2.2, Exercises 1, 3, and 7 — odd-numbered, answers in the back.

Coming up

Homework 5 is due Friday, September 18, 11:59 p.m.

Questions I still have